

Cerebral Hemodynamics in Pediatric Abusive Head Trauma: 3 Severe Cases with Preserved Motor Cortex, Hyperperfusion, and Recovery of Mild Paralysis

Mitsuru TAMURA,¹ Shinji YAMASHITA,¹ Tomoki KAWANO,¹ Satoru KOMAKI,¹
Takeru TSUKINO,² Koutarou KOJIMA,² Kenichi MAEDA,² Yasuhiro KIMOTO,²
Yoshihito KADOTA,³ Minako AZUMA,³ and Yoshiko OKITA¹

¹Department of Neurosurgery, Faculty of Medicine, University of Miyazaki, Kiyotake, Miyazaki, Japan

²Department of Pediatrics, Faculty of Medicine, University of Miyazaki, Kiyotake, Miyazaki, Japan

³Department of Radiology, Faculty of Medicine, University of Miyazaki, Kiyotake, Miyazaki, Japan

Abstract

Abusive head trauma in infants and young children can have a significant impact on neurological outcomes and, in severe cases, may be life-threatening. We report 3 cases of abusive head trauma that presented with acute subdural hematomas on computed tomography scans, accompanied by extensive low-density areas and parenchymal brain swelling. All patients exhibited impaired consciousness due to brain injury and underwent craniotomy for hematoma evacuation as well as extensive decompressive craniectomy. Despite the severity of the initial presentation, hemiparesis was mild and gradually improved over several months. Postoperative magnetic resonance imaging revealed widespread parenchymal injury but preservation of the corticospinal tract, including the precentral gyrus. In the acute phase, diffusion-weighted imaging showed no irreversible infarction in the motor cortex, and arterial spin labeling demonstrated increased perfusion in peri-motor regions of the affected hemisphere. These findings suggest that preserved corticospinal pathways and compensatory hyperperfusion may correlate with favorable motor recovery even in the presence of extensive parenchymal damage. These cases highlight the radiological features and short-term neurological outcomes of abusive head trauma, demonstrating preserved motor function despite extensive parenchymal damage.

Keywords: abusive head trauma, motor function, motor cortex, regional blood flow

Introduction

Abusive head trauma (AHT) is defined as cranial and intracranial injuries in children under 5 years of age caused by blunt force trauma, violent shaking, or a combination of both, inflicted intentionally.¹⁾ Symptoms of AHT vary depending on the severity of the trauma. In infants, atypical symptoms such as lethargy, irritability, poor feeding, vomiting, altered consciousness, and seizures are frequently observed.^{2,3)} The pathophysiology of AHT involves direct traumatic impact to the head and damage to the brain parenchyma or blood vessels due to shear forces from rotational acceleration and deceleration. This mechanism is particu-

larly associated with subdural hematomas, subarachnoid hemorrhages, diffuse axonal injury, and retinal hemorrhages. Moreover, AHT is linked to worse outcomes than general head trauma, with approximately half of affected children developing neurological sequelae, including severe motor deficits.

Brain damage resulting from AHT is heterogeneous, making prognosis difficult to predict. Magnetic resonance imaging (MRI) is widely used to assess the degree and nature of the injury, particularly in acute settings. Diffusion tensor imaging allows for the visualization of ischemic foci that may be missed by computed tomography (CT).⁴⁾ Hypoxic-ischemic injuries in AHT, characterized by hyper-

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intensity on diffusion-weighted imaging (DWI), are the most common type of parenchymal damage in victims and are correlated with more severe outcomes.⁵⁾ However, the precise mechanisms underlying AHT-related brain damage remain unclear. An effective radiological assessment that reliably predicts motor deficits has not yet been established.

Arterial spin labeling (ASL) has emerged as a non-invasive method for assessing cerebral perfusion by directly measuring cerebral blood flow (CBF) without the need for contrast agents. ASL signal intensity correlates with CBF, which often reflects the metabolic demand of brain tissue. Other perfusion studies, such as those based on the mean transit time (MTT), provide complementary information.⁶⁾ For example, reduced CBF or prolonged MTT are markers of perfusion delay or ischemia in many traumatic brain injury (TBI) settings.⁷⁾ In prior reports on TBI, imaging has shown decreased CBF and cerebral blood volume, and prolonged MTT, consistent with hypoperfusion.^{8,9)} These findings generally predict poorer outcomes. In contrast, in the cases reported in this study, ASL demonstrated increased signal intensity in the affected hemisphere, particularly in the peri-motor cortex. Although such hyperperfusion is usually interpreted as abnormal, we considered it to reflect a compensatory hemodynamic response rather than pathological hyperemia. Increased CBF may accompany elevated metabolic demand in viable but vulnerable tissue in the traumatic brain.¹⁰⁾ In our cases, the preserved corticospinal tract on DWI combined with ASL hyperperfusion may explain why hemiparesis was mild and improved over time despite extensive parenchymal injury. Thus, ASL hyperperfusion could serve as a potential radiological marker of preserved motor function and a favorable neurological outcome.

This study retrospectively examines cases of AHT that were surgically treated at our institution, focusing on their radiological characteristics and neurological outcomes, particularly the preservation of motor function despite severe parenchymal damage.

ASL Imaging

ASL was performed using a pseudocontinuous labeling scheme with the following parameters: repetition time of 6,000 ms in Cases 1 and 2, and 4,284.4 ms in Case 3; echo time of 21.7 ms in Cases 1 and 2, and 10.7 ms in Case 3. Post-labeling delay (PLD) was set to 1,800 ms in Cases 1 and 2, and 2,000 ms in Case 3. A three-dimensional background-suppressed readout was used to minimize static tissue signals.

Post-labeling delay was adjusted according to body size and circulation time. Specifically, for infants and young children, a shorter PLD of 1,500-1,800 ms is commonly applied to account for their faster arterial transit, whereas in older children and adults, a PLD of 1,800-2,000 ms is used

to better capture capillary-phase perfusion.¹¹⁾

Case Reports

Case 1

A 3-month-old girl was brought to the emergency department with vomiting, facial pallor, and impaired consciousness. Head CT at the referring hospital revealed a left acute subdural hematoma (ASDH), and she was transferred to our institution for further management. On arrival, her Glasgow Coma Scale (GCS) score was E1V3M5, and anisocoria was observed. Fundoscopic examination revealed bilateral retinal hemorrhages, raising suspicion of AHT. Head CT demonstrated a thin left-sided ASDH and a low-density area in the left cerebral hemisphere (Figure 1 a). The anterior fontanelle was tense, suggesting elevated intracranial pressure (ICP). Emergency craniotomy for hematoma evacuation, extensive decompressive craniectomy, and ICP monitor placement were performed. Postoperatively, right hemiparesis was observed. ICP was well controlled with medication, leading to an improvement in consciousness (GCS E4V4M6) and right hemiparesis (manual muscle testing 4/5) by postoperative day 3, with near-complete recovery by 3 months. MRI performed 11 days after the injury revealed extensive damage in the left cerebral hemisphere; however, the corticospinal tract remained anatomically intact (Figure 1b). DWI did not show high signal intensity in the injured area (Figure 1c). ASL demonstrated increased perfusion in the left hemisphere (Figure 1 d), and magnetic resonance angiography (MRA) revealed more prominent cortical vessels extending peripherally on the affected side than on the contralateral side (Figure 1e). MRI examination was performed in the absence of seizure activity.

Case 2

A 3-month-old boy was initially transported to a referring hospital due to impaired consciousness, facial pallor, and right-sided facial seizures. Head CT revealed a left ASDH, and he was subsequently transferred to our hospital. Upon arrival, his GCS score was E1VTM4, and seizures had resolved. Fundoscopic examination revealed bilateral retinal hemorrhages, suggesting AHT. CT imaging showed a thin left-sided ASDH and a low-density area in the left cerebral hemisphere (Figure 2a). The anterior fontanelle was tense, indicating elevated ICP. Emergency craniotomy for hematoma evacuation, extensive decompressive craniectomy, and ICP monitor placement were performed. Postoperatively, right hemiparesis was noted. ICP was effectively controlled with medical management, resulting in improved consciousness (GCS E4V4M6) and near-complete resolution of hemiparesis within 1 month. MRI performed 10 days after the injury demonstrated extensive damage in the left cerebral hemisphere; however, the corticospinal tract remained anatomically preserved (Figure 2b). DWI

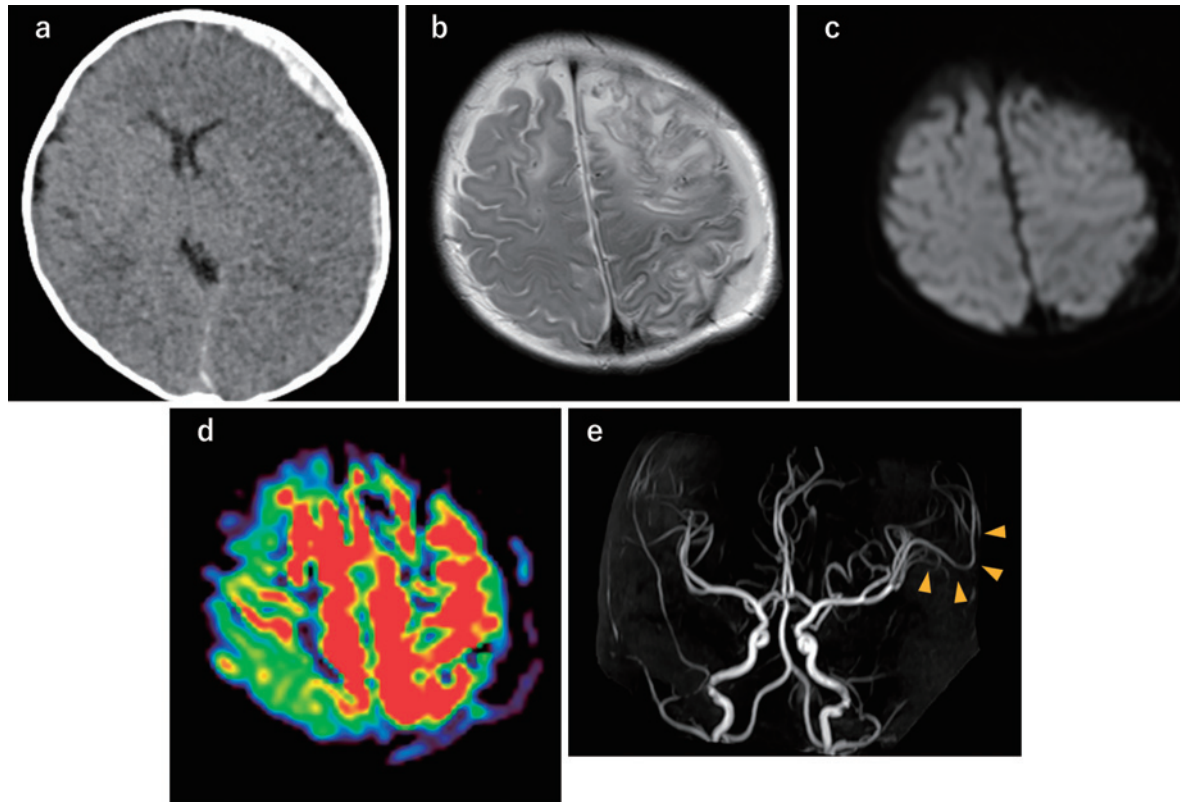


Figure 1 Preoperative head CT (a), postoperative MRI (b, c, and d), and MRA (e) performed 11 days after surgery.

- a. A thin left acute subdural hematoma with widespread blurring of the gray-white matter junction in the left cerebral hemisphere and a midline shift.
- b. Although the left cerebral hemisphere sustained extensive damage, the precentral gyrus was preserved on T2-weighted imaging.
- c. DWI did not reveal any high signal intensity at the site of injury.
- d. ASL demonstrated increased signal intensity in the left cerebral hemisphere, including the precentral gyrus.
- e. The signal intensity of the left middle cerebral artery, which perfuses the left cerebral hemisphere, was enhanced compared to the contralateral side (orange arrowhead).

ASL: arterial spin labeling; CT: computed tomography; DWI: diffusion-weighted imaging; MRA: magnetic resonance angiography; MRI: magnetic resonance imaging

showed mild hyperintense lesions in the right occipital and left frontal lobes, sparing the motor cortex (Figure 2c). ASL revealed increased perfusion in the left cerebral hemisphere (Figure 2d). MRA demonstrated more prominent cortical vessels extending peripherally on the affected side compared to the contralateral hemisphere (Figure 2e). MRI was performed without any clinically observed seizures.

Case 3

A 2-year-old girl was brought to our hospital by emergency transport due to impaired consciousness and seizures. On arrival, her GCS score was E1V1M4, and anisocoria was observed. Fundoscopic examination revealed bilateral retinal hemorrhages, raising suspicion of AHT. Head CT demonstrated a right ASDH, extensive blurring of the gray-white matter junction in the right cerebral hemisphere, and a midline shift (Figure 3a). Emergency craniotomy for hematoma evacuation, extensive decompressive craniectomy, and ICP monitor placement were performed.

Despite ongoing medical management, bilateral pupil dilation occurred on postoperative day 5, and ICP increased to 35-40 mmHg. Follow-up CT revealed extensive low-density areas and brain swelling in the right cerebral hemisphere. MRI showed widespread high signal intensity on DWI in the right cerebral cortex and left frontal lobe (Figure 3b), prompting additional internal decompression via right temporal lobectomy. Postoperatively, left hemiparesis was observed. MRI performed 3 weeks after the injury revealed extensive damage in the right cerebral hemisphere; however, the corticospinal tract, including the precentral gyrus, remained anatomically preserved (Figure 3c). Mild hyperintensity on DWI was also observed in the motor cortex of the right hemisphere and the left frontal lobe 3 weeks after the injury (Figure 3d), similar to the findings on pre-emergency craniotomy DWI (Figure 3b). However, the signal intensity appeared fainter compared to that observed on the pre-emergency craniotomy DWI. ASL imaging demonstrated increased perfusion in the motor cortex (Figure

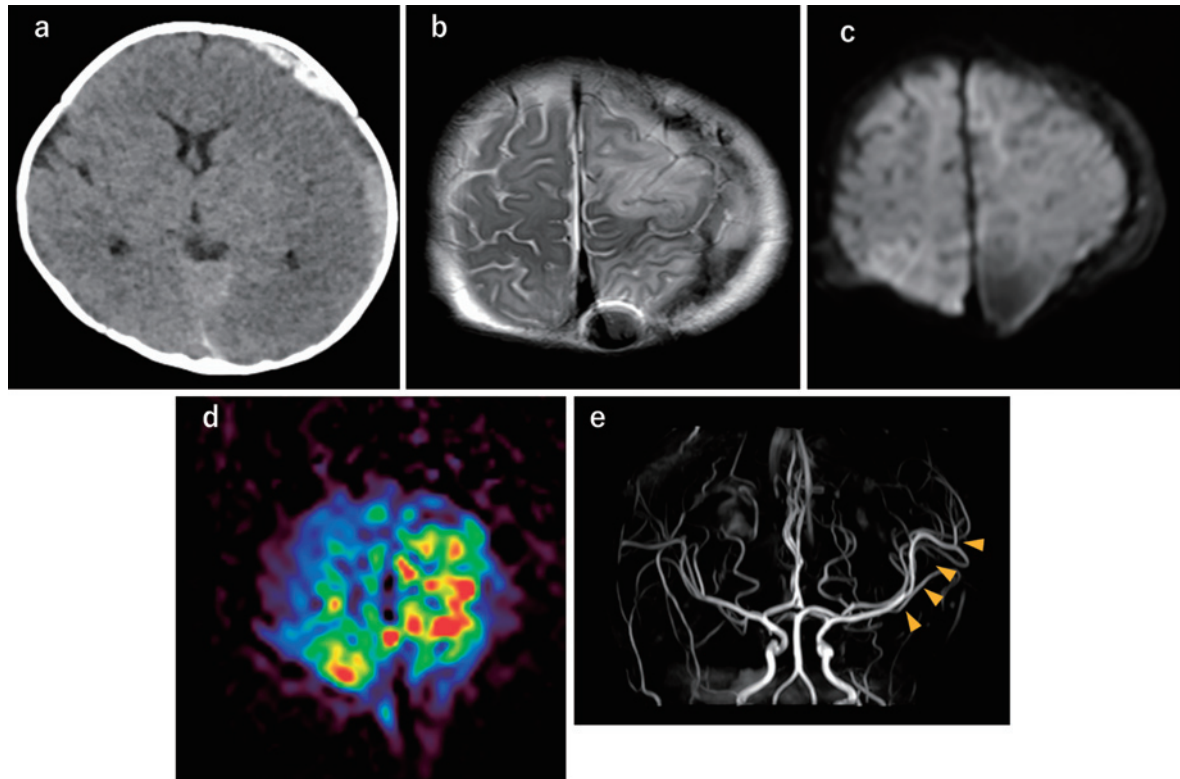


Figure 2 Preoperative head CT (a) and postoperative MRI (b, c, and d) and MRA (e) performed 10 days after surgery.

- a. A thin left acute subdural hematoma with widespread blurring of the gray-white matter junction in the left cerebral hemisphere and a midline shift.
- b. Although the left cerebral hemisphere sustained extensive damage, the precentral gyrus was preserved on T2-weighted imaging.
- c. DWI showed high signal intensity in the bilateral frontal and occipital lobes. The signal spread is asymmetrical.
- d. ASL demonstrated increased signal intensity in the left cerebral hemisphere, including the precentral gyrus.
- e. The signal intensity of the left middle cerebral artery, which perfuses the left cerebral hemisphere, was enhanced compared to the contralateral side (orange arrowhead).

ASL: arterial spin labeling; CT: computed tomography; DWI: diffusion-weighted imaging; MRA: magnetic resonance angiography; MRI: magnetic resonance imaging

3e). No clinical seizures were observed during MRI. Her level of consciousness improved to GCS E4V4M6, and partial recovery of left hemiparesis was noted at 3 months. Moderate spastic paresis persisted at 6 months.

Background of abuse diagnosis

In all 3 cases, the exact mechanism of injury was unknown. No caregivers reported trauma, and no scalp lacerations or skull fractures were observed. Cases 2 and 3 had body bruises. CT scans showed no fractures in the skull, extremities, or trunk, but all 3 cases exhibited bilateral retinal hemorrhages, subdural hematomas, and cerebral swelling, suggesting non-accidental injury, including shaken baby syndrome. Suspicion of abuse was supported by inconsistencies in caregiver explanations, body surface findings, and evaluation by the Child Protection Team at University of Miyazaki Hospital. Nutritional status was normal in Cases 1 and 2, whereas Case 3 had poor weight gain over the preceding 6 months. No additional fractures

were detected. Based on these findings, abuse was suspected, and the children were placed under temporary protective custody after notification to the child guidance center and police.

Discussion

Despite extensive brain injury and atrophy in the affected cerebral hemisphere on short-term follow-up MRI, the corticospinal tract remained structurally intact in all cases. This preservation likely contributed to mild hemiparesis and favorable short-term functional outcomes. Previous studies on pediatric TBI suggest that there may be regional differences in the brain's reparative processes or that recovery related to brain plasticity may vary by region.^{12,13} In our series, DWI demonstrated minimal ischemic changes in the motor cortex compared to other affected regions, which may explain the reversibility of motor deficits.

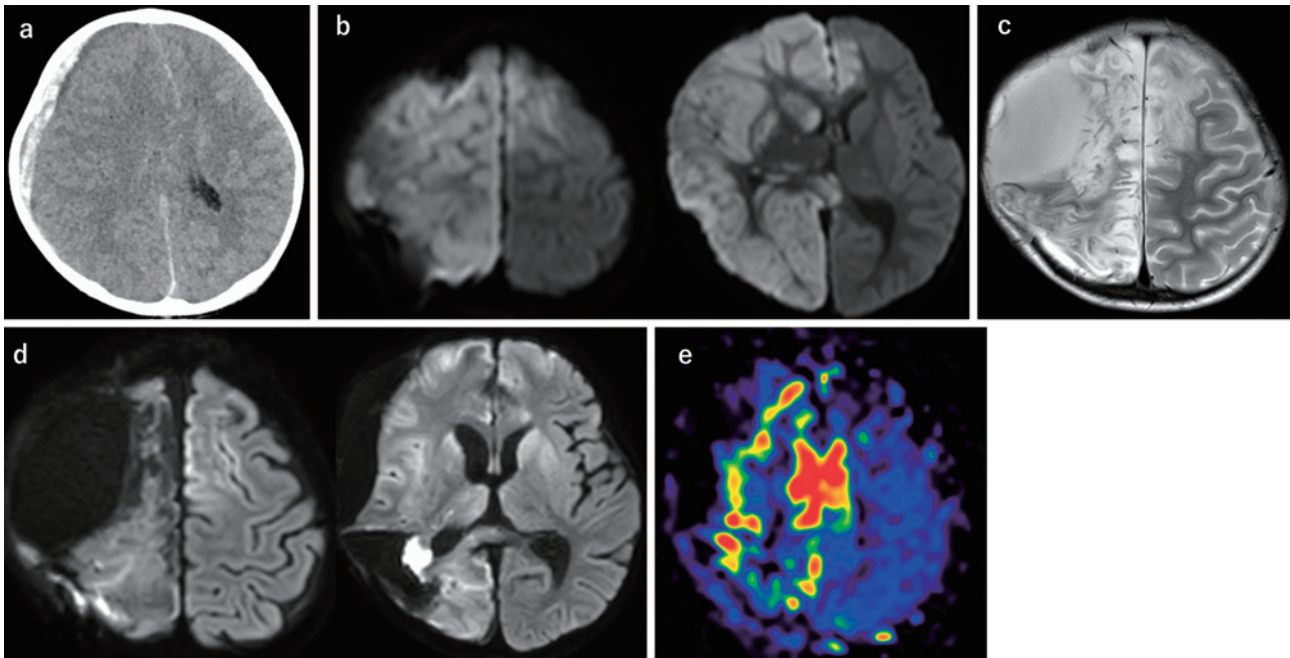


Figure 3 Preoperative head CT (a), pre-emergency craniotomy DWI (b), and postoperative MRI performed 3 weeks after surgery (c, d, and e).

- a. A thin right acute subdural hematoma with widespread blurring of the gray-white matter junction in the right cerebral hemisphere and a midline shift.
- b. DWI showed high signal intensity in the right cerebral cortical hemisphere and left frontal cortex.
- c. Although the right cerebral hemisphere sustained extensive damage, the precentral gyrus was preserved on T2-weighted imaging.
- d. DWI showed that much of the area previously exhibiting high signal intensity remained as faintly hyperintense regions.
- e. ASL demonstrated increased signal intensity in the right cerebral hemisphere, including the precentral gyrus.

ASL: arterial spin labeling; CT: computed tomography; DWI: diffusion-weighted imaging; MRA: magnetic resonance angiography; MRI: magnetic resonance imaging

Cerebral ischemia following head trauma can arise through several distinct mechanisms: (1) global hypoxic-ischemic injury due to a mismatch between CBF and metabolic demand, (2) focal ischemia caused by reduced perfusion pressure from local mass effects or vascular compression, and (3) territorial infarction resulting from vascular compromise associated with brain herniation. Hypoxic-ischemic injury, characterized by hyperintensity on DWI, is the most common type of parenchymal damage observed in AHT and is typically associated with more severe neurological outcomes.⁵⁾ This mechanism may also be involved in the development of cerebral infarctions following ASDH in pediatric patients.¹⁴⁾ In cases of ASDH with significant mass effect, cerebral infarctions may result from compression of the anterior cerebral artery against the falx cerebri and the posterior cerebral artery against the tentorium cerebelli. In contrast, other reports have indicated that cerebrovascular disturbances in AHT are often bilateral and multifocal, involving multiple vascular territories.¹⁵⁾ In our series, Cases 1 and 2 were the most consistent with mechanism.¹⁾ Case 1 showed extensive hemispheric damage without DWI hyperintensity but with in-

creased ASL perfusion and prominent cortical vessels on MRA. These findings suggested circulatory-metabolic mismatch rather than territorial ischemia. Similarly, Case 2 demonstrated bilateral but asymmetric DWI hyperintensity despite well-controlled postoperative ICP. This finding further supported the role of global cerebral hypoxia, likely induced by seizures, bradycardia, or apnea, rather than arterial compression or perfusion failure. In contrast, Case 3 was consistent with mechanism.³⁾ In addition to ischemic changes in the middle cerebral artery territory on the affected side—where compression from the ASDH was most pronounced—ischemia was also observed in the posterior cerebral artery territory on the same side and in the bilateral anterior cerebral artery territories. On follow-up MRI, these DWI lesions had progressed to complete infarction, consistent with arterial compression due to elevated ICP and brain herniation. Taken together, these findings highlight the heterogeneity of ischemic mechanisms in AHT, ranging from global hypoxic-ischemic injury due to circulatory-metabolic mismatch (Cases 1 and 2) to territorial infarction resulting from herniation-induced vascular compression (Case 3).

In Cases 1 and 2, no high signal intensity was observed on DWI in the motor cortex. Moreover, although Case 3 was more severe, the high signal intensity in the motor cortex was relatively limited compared to other affected regions. These findings suggest that the motor cortex may have sustained only minimal ischemic damage, potentially contributing to the preservation of motor function in all 3 cases. In contrast, previous reports have described the “big black brain” phenomenon in infants with AHT. This phenomenon is characterized by hemispheric hypodensity and progressive diffuse cortical encephalomalacia. Such cases often demonstrate clinicoradiological dissociation, with clinical improvement appearing transiently despite severe radiological progression, and are typically associated with poor prognosis.¹⁶⁾ It is well established that hypoxic-ischemic injury (mechanism 1) preferentially involves brain regions with high metabolic demand. Vulnerability is also age-dependent: in a 2-year-old child, cortical layers III, V, and VI and the hippocampus are particularly susceptible, whereas in infants younger than 1 year, regions undergoing active myelination—such as the motor cortex and corticospinal tract at 3 months of age—are particularly vulnerable. The preservation of the motor cortex in Cases 1 and 2, despite global hypoxic-ischemic injury, was thus an unexpected finding. The increased ASL signal observed in these cases, which might superficially be interpreted as pathological hyperperfusion, may in fact represent a compensatory hemodynamic response aimed at maintaining perfusion in threatened but viable tissue. However, even under compensatory hyperperfusion, blood flow may remain insufficient given the heightened metabolic demand, thereby creating a state analogous to ischemic penumbra. This pathophysiological condition could explain the presence of transient hemiparesis and its subsequent recovery in our patients. Previous studies have reported that increases in cerebral perfusion are often accompanied by reductions in venous oxygen saturation. This phenomenon may reflect a neuroprotective compensatory mechanism, wherein CBF is increased to meet the heightened metabolic demands following head trauma.¹⁰⁾ In our cases, MRA demonstrated improved visualization of cortical branches on the surgical side in 2 patients, suggesting compensatory development of leptomeningeal anastomoses to maintain cerebral perfusion. Notably, the corticospinal tract receives blood supply from both penetrating and cortical branches of the cerebral arteries, a dual vascular supply that may enhance its resilience to ischemic injury by ensuring adequate perfusion.

In contrast, in the severe case (Case 3), MRA signal intensity did not increase on the surgical side, and large areas of high signal intensity were observed on DWI, indicating widespread ischemia. These findings were accompanied by corresponding high signal intensity on T2-weighted images. In severe cases with critically elevated ICP requiring internal decompression, perfusion of the cor-

tical branches can be markedly impaired, potentially leading to irreversible ischemic damage. In this study, MRA and ASL were used to assess changes in CBF following AHT. However, due to the severity of the injuries, the timing of imaging evaluations was not standardized, which precluded longitudinal assessment. Furthermore, the imaging findings reported here reflect only short-term neurological status; thus, further studies incorporating long-term outcomes are warranted. Several limitations should be acknowledged. First, metabolic activity was not directly evaluated in this study. Therefore, we cannot confirm whether the apparent hyperperfusion observed on ASL truly reflected increased CBF or instead represented a state of relative ischemia due to circulatory-metabolic mismatch. Second, alternative explanations such as hyperemia secondary to vasoparalysis cannot be excluded. This mechanism would be consistent with both the transient hemiparesis and the development of vasogenic edema observed in some of our cases. Future investigations combining perfusion imaging and metabolic assessment are required to clarify these pathophysiological processes.

Conclusions

In cases of significant parenchymal brain injury in patients with AHT, compensatory neuroprotective responses may lead to increased CBF, which could help preserve the corticospinal tract and maintain motor function.

Conflicts of Interest Disclosure

All authors have no conflict of interest.

Ethics Approval

This research was approved by the Ethics Committee of Miyazaki University (approval no.: C-0180).

Informed Consent

Informed consent from the parents could not be obtained due to the abusive nature of these pediatric cases; however, the study protocol was reviewed and approved by the institutional ethics committee (approval no.:C-0180).

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Corresponding author: Mitsuru Tamura, MD

Department of Neurosurgery, Faculty of Medicine, University of Miyazaki, 5200 Kihara, Kiyotake, Miyazaki 889-1692, Japan.

e-mail: mitsuru_tamura@med.miyazaki-u.ac.jp